INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Business Analytics

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____



Signature of Student

Seal & Signature of IPO

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Business Analytics successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample data and case studies, and the supervisor's patient and detailed feedback helped me understand how business data is collected, analysed and turned into decisions.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Business Analytics is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

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PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Business Analytics at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Every business produces data all the time - each bill, each order, each enquiry and each payment is a small piece of information. A shopkeeper who knows which items sell best in which month, which customers return and where money is lost can take better decisions than one who depends only on memory and guesswork. Business analytics is the practice of collecting such data, cleaning it, analysing it with statistics and tools such as MS Excel, presenting it in charts and dashboards, and using the findings to improve the business. It is one of the most in-demand skills today, because organisations of every size are looking for people who can turn data into insight. This report describes how the main steps of business analytics were studied and how a sales performance analysis and a KPI dashboard were built for a fictitious retail store using dummy data.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that it gives a clear and honest account of the internship experience. All examples in this report use dummy data and fictitious names; no real personal or business information has been handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Business Analytics at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Business analytics is the use of data, statistical methods and tools to understand business performance, find the reasons for results, predict what may happen and support decisions. It is commonly divided into four kinds. Descriptive analytics says what happened (sales last month), diagnostic analytics explains why it happened (sales fell because of a stock-out), predictive analytics estimates what is likely to happen (next month's demand) and prescriptive analytics suggests what to do (how much to order). The tool most widely used by beginners and by small businesses is MS Excel, with its formulas, pivot tables and charts.

Objectives. The internship aimed to (i) understand the types and process of business analytics, (ii) use MS Excel for business analysis, (iii) collect and clean data, (iv) calculate and interpret descriptive statistics, (v) design KPIs aligned with business goals, (vi) choose and build charts, (vii) learn basic forecasting, (viii) build a dashboard, (ix) analyse a business case, and (x) present findings and recommendations to stakeholders.

Work carried out. Each topic had a concept session followed by hands-on analysis in MS Excel and, for charts, in a plotting tool, using a dummy twelve-month data set of a fictitious retail store, Mahadev Mart. The student cleaned the data, calculated statistics, built pivot-style summaries, designed KPIs, drew charts, fitted a trend line, built a dashboard and wrote recommendations. The later part of the programme was devoted to the project, a case study exercise, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled **Business Performance Analysis & KPI Dashboard**. The project analysed twelve months of dummy sales, cost and customer data of Mahadev Mart, calculated its main statistics and KPIs, found the strong and weak categories and months, forecast the next month by a trend line and built a one-page KPI dashboard with recommendations.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of the types and process of analytics, practical skill in Excel formulas, summaries and charts, data cleaning, descriptive statistics, KPI design, simple forecasting, dashboard building and presentation of insights, and the habit of backing every statement with numbers.

Key observations. Good analysis starts with a clear business question and not with the data; clean data matters more than clever methods; averages can hide big differences between categories and months; a chart that is simple and honest is worth more than many complicated ones; and the value of analysis lies in the action it leads to.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Business Analytics

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand the types and process of business analytics (Week 1).
2. Use MS Excel for business analysis - formulas, pivot tables and charts (Week 1).
3. Collect and clean data from different sources (Week 1-2).
4. Calculate and interpret descriptive statistics (Week 2).
5. Design KPIs and visualise data with suitable charts (Week 2-3).
6. Learn forecasting basics and build a dashboard (Week 3-4).
7. Analyse a business case, present insights and complete the project (Weeks 4-5).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the idea, its main terms and how it is used in real organisations. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - cleaning a messy data set, calculating mean, median, standard deviation and percentages, building pivot-style summaries with SUMIFS and COUNTIFS, designing KPIs, drawing bar, line and scatter charts, fitting a trend line, building a dashboard and writing the findings in plain words, all in MS Excel. Only sample data and fictitious examples were used, so that no real personal or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, files and screenshots were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Business Analytics - Concepts Studied

3.1 What is business analytics?

Business analytics converts the facts recorded in a business into knowledge that can be used to take decisions. The need for it has grown because businesses now record more data than ever - sales in billing software, payments in mobile apps, enquiries on social media - and because customers, competitors and costs change quickly. The four kinds of analytics form a ladder of increasing value: descriptive, diagnostic, predictive and prescriptive. Most small businesses can gain a great deal from the first two kinds with simple tools, and even a little prediction makes planning better.

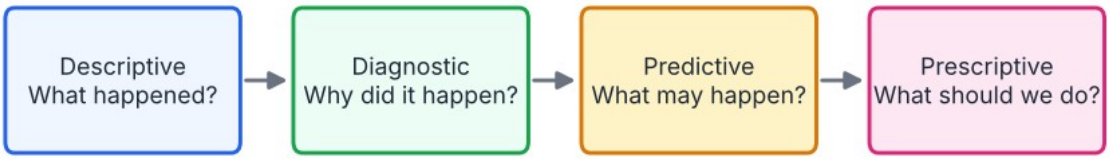


Figure 1: The four kinds of business analytics

The usual process of an analytics project has six steps: define the business question, collect the data, clean the data, analyse it, visualise the results and recommend an action. The most important step is the first, because a vague question such as 'analyse our sales' leads to a vague answer, while a clear question such as 'which categories made the most profit last year, and why did the margin fall in February?' leads to a useful one.



Figure 2: The process of a business analytics project

3.2 MS Excel for business analysis

MS Excel is the most widely used tool for business analysis because it combines a table for the data, a calculator for formulas and a chart maker. The functions used most often are SUM, AVERAGE, MEDIAN, MIN, MAX, COUNT, IF, SUMIF and SUMIFS (sums by a condition), COUNTIFS, VLOOKUP or XLOOKUP (to look up a value in another table), TEXT and DATE functions, and ROUND. A PivotTable summarises a large table by categories without a formula: dragging 'Category' to the rows and 'Sales' to the values immediately gives sales by category. Good practice includes keeping the raw data in one sheet and the analysis in other sheets, using one row per record and one column per field, naming the sheets and never typing over a formula. The formulas used in the project are shown below.

Excel formulas used in the analysis

Total sales: =SUM(B2:B13)
Average monthly sales: =AVERAGE(B2:B13)
Profit of a month: =B2-C2
Profit margin %: =(B2-C2)/B2*100
Sales of one category: =SUMIFS(Sales,Category,"Clothing")
Share of a category: =SUMIFS(Sales,Category,"Clothing")/SUM(Sales)
Month-on-month growth %: =(B3-B2)/B2*100
Trend forecast for month 13: =FORECAST.LINEAR(13,B2:B13,A2:A13)

3.3 Data collection and cleaning

Data comes from internal sources - billing and accounting software, ERP and CRM systems, stock registers and employee records - and from external sources such as government statistics, industry reports, surveys and social media. Before analysis the data must be cleaned, because real data is always imperfect. The usual problems and their treatment are shown in the table below. A rule learnt in the internship is never to change the original data file: work on a copy, and keep a note of every change made so that the work can be checked.

Problem	Example	Treatment
Missing values	Customer count blank for one day	Fill from another source, use the average, or leave out with a note
Duplicates	The same bill entered twice	Remove using the key (bill number)
Wrong format	Dates as text; 'Rs. 1,200' as text	Convert to date and number formats
Inconsistent spelling	'Grocery', 'grocery', 'Grocry'	Standardise names; use a drop-down list

Problem	Example	Treatment
Outliers	A sale of Rs. 5,00,000 in a day of Rs. 15,000 average	Check against the bill; correct, keep or remove, with a reason
Wrong units	Some figures in rupees and some in thousands	Convert all to one unit

3.4 Descriptive statistics

Descriptive statistics summarises a set of numbers with a few figures. The measures of central tendency tell the typical value: the mean (sum divided by count), the median (the middle value when sorted) and the mode (the most frequent value). The measures of spread tell how much the values differ: the range (maximum minus minimum), the standard deviation (the typical distance from the mean) and the coefficient of variation (standard deviation divided by mean, as a percentage). Quartiles divide the sorted data into four parts. For the monthly sales of the dummy store, in Rs. thousand, the main measures are shown below.

Measure	Value	Meaning
Mean	378.3	Average monthly sales
Median	357.5	Middle value; lower than the mean because of the high months (Nov, Dec)
Minimum / Maximum	295 (Feb) / 520 (Nov)	Lowest and highest month
Range	225	Maximum minus minimum
Standard deviation	67.9	Typical distance from the mean
Coefficient of variation	18.0%	Spread relative to the mean; moderate seasonality
First and third quartile	337.5 and 403.75	Half of the months fall between these values

The mean is higher than the median because a few large months pull it up. This is a reminder that the average alone can mislead: for planning stock in a normal month, the median of Rs. 357.5 thousand is a better guide than the mean, while for planning the festival season the high months must be treated separately.

3.5 KPI design

A Key Performance Indicator (KPI) is a measure that shows how well the business is doing against an important goal. A good KPI is relevant to a goal, measurable from available data, clearly defined, comparable over time or with a target and owned by a person. The number of KPIs should be small - five to ten for a business unit - and they should be a mix of leading indicators, which predict results (for example, the number of customers visiting), and lagging indicators, which report results (for example, profit). The goals and KPIs designed for the store are shown below.

Goal	KPI	Formula	Value for the year
Grow sales	Sales growth, second half over first half	(H2 sales - H1 sales) / H1 sales	29.3% (Rs. 25.6 lakh against Rs. 19.8 lakh)
Earn profit	Net profit margin	Profit / sales x 100	26.5%
Attract customers	Customers per month	Count of bills	616 per month on average
Raise spending	Average bill value	Sales / number of customers	Rs. 615 (Rs. 642 in November)
Use the best categories	Profit share of the top category	Category profit / total profit	Grocery 34.8%

Goal	KPI	Formula	Value for the year
Control cost	Cost as a percentage of sales	Cost / sales x 100	73.5%

3.6 Data visualisation

A chart is meant to help the reader see a pattern at once. The type of chart depends on the question: a line chart shows change over time, a bar chart compares categories, a scatter plot shows the relationship between two numbers, a histogram shows the distribution of values, and a pie chart shows shares of a whole (only when there are few parts). Good charts have a clear title that states the message, labelled axes with units, a limited number of colours used with a purpose, no unnecessary decoration and a baseline of zero for bar charts. Three-dimensional effects, too many colours and truncated axes mislead the reader and were avoided.

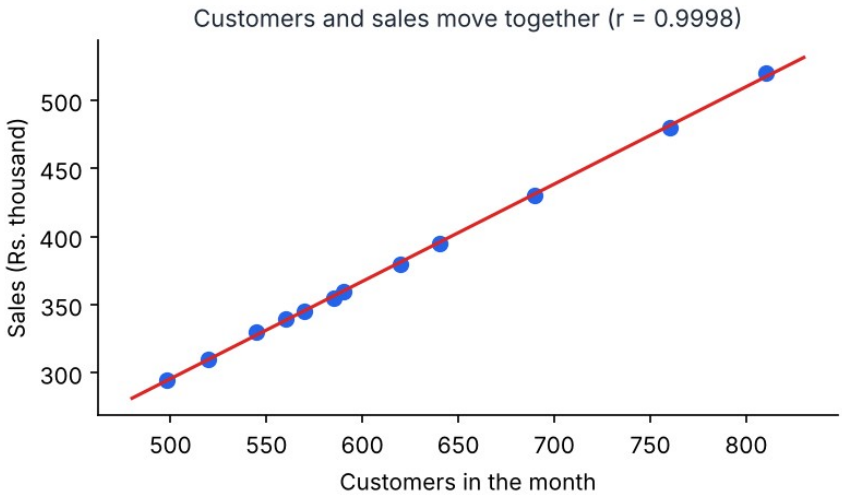


Figure 3: Scatter plot of customers against sales, with a trend line

The scatter plot shows that the sales of the store rise almost exactly in step with the number of customers, with a correlation coefficient of 0.9998. This tells the owner that the most reliable way to raise sales is to bring in more customers, while raising the average bill value (which moved only between Rs. 592 and Rs. 642) is a smaller lever. The student noted that correlation shows a relationship but does not prove that one thing causes the other.

3.7 Forecasting basics

Forecasting estimates future values from past data. The simple methods studied were the moving average, the linear trend line and the use of seasonal indices. A linear trend line is fitted to the points by the method of least squares, and its equation is used to estimate the next value. For the twelve months of sales, Excel's trend line gave: sales = 266.5 + 17.2 x month number, with an R-squared of 0.834, which means that the trend explains about 83% of the variation in sales. The trend value for the thirteenth month is 266.5 + (17.2 x 13) = about Rs. 490 thousand.

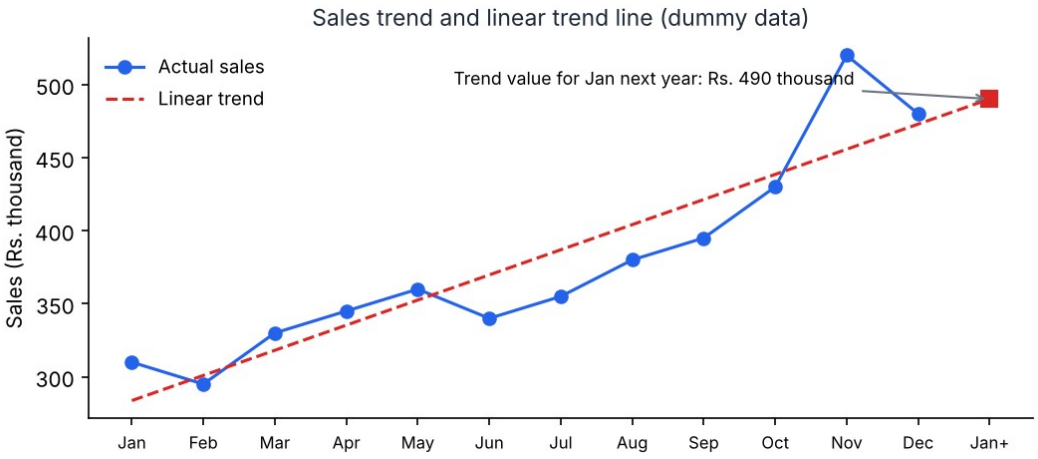


Figure 4: Monthly sales and the linear trend line

The result must be used with care. The trend line ignores the seasonality of the business - the festival months of October to December are well above the line, and the months of February and June are below it - so the forecast for January, which is usually a slow month after the festival, is probably too high. A better method would adjust the trend with a seasonal index for each month, and the forecast should always be compared with the actual result and revised. The student therefore presented the forecast as a range and not as a single number.

3.8 Building a dashboard

A dashboard is a single page that shows the most important KPIs and charts so that a manager can understand the position in a few seconds. A good dashboard begins with the main KPIs in large numbers at the top, followed by a few charts that answer the main questions, arranged in the order of the reader's attention. It should fit on one screen or one page, use consistent colours and labels, highlight what needs attention (for example, a margin that has fallen below target) and be updated regularly. The dashboard built for the store is shown below; it was prepared from the Excel summaries and drawn with a plotting tool.

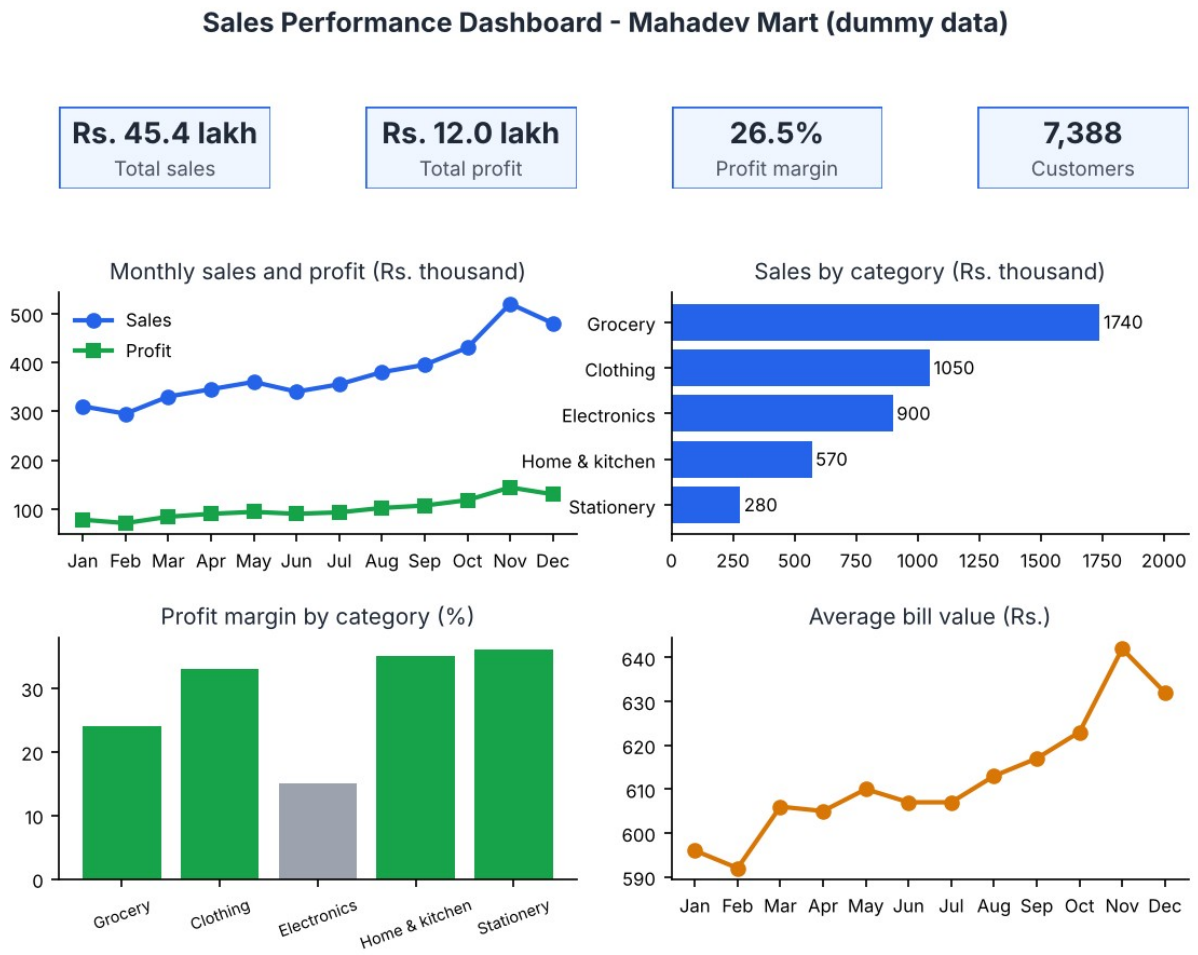


Figure 5: KPI dashboard of the dummy retail store

The dashboard shows at a glance that the year's sales were Rs. 45.4 lakh with a profit of Rs. 12.0 lakh (a margin of 26.5%) and 7,388 customers; that sales rose strongly in the second half and peaked in November; that grocery brings the most sales but electronics has the lowest margin at 15%; and that the average bill value has risen steadily through the year.

3.9 Business case analysis

A business case analysis applies analytics to a real problem. The steps are to state the problem and the decision to be taken, to list the alternatives, to collect and analyse the data for each alternative, to compare them on cost, benefit and risk, and to recommend one with the reasons and the next steps. A useful tool is a simple table that compares the options on the measures that matter, with the assumptions written down. The student practised on the case of whether the store should add a second branch, expand the clothing section or start home delivery, using the profit data of the categories (section 10).

3.10 Presenting insights to stakeholders

An analysis has value only when the decision-maker understands it and acts on it. A good presentation of findings begins with the answer, not with the method: 'Clothing and home items give 50% of the profit from

36% of the sales, so give them more space.' It then shows two or three charts that support the answer, explains the main assumptions and limits, and ends with clear recommendations and the next step. The language should be simple, with no jargon, and every number should have its unit and its comparison (against last year, the target or another category). The student prepared a one-page summary and a five-slide presentation of the store analysis and presented it to the supervisor.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, business analytics overview, MS Excel for business analysis and data collection and cleaning, each with a study session and a practice task.
Week 2	Days 7-12	Measures and charts: descriptive statistics, KPI design and data visualisation.
Week 3	Days 13-18	Forecasting and dashboards: forecasting basics and dashboard building.
Week 4	Days 19-24	Application: business case analysis, insight presentation, the Business Performance Analysis & KPI Dashboard project, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student studied the types and process of business analytics, practised the main Excel functions and PivotTables, and cleaned a messy data set by removing duplicates and correcting formats.

Week 2: In the second week the student calculated descriptive statistics for the twelve months of sales, designed KPIs for six business goals and drew line, bar and scatter charts, learning how to choose the right chart for each question.

Week 3: The third week was devoted to forecasting with a trend line and to building the KPI dashboard from the Excel summaries.

Week 4: In the fourth week the student analysed business cases, prepared a presentation of the findings, completed the project, a case study exercise and a mock practical exercise.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Business Analytics Overview

Concept: Descriptive, predictive and prescriptive analytics in business. **Practice carried out:** Writing three examples of each kind of analytics for a retail store, a school and a bank.

6.2 Excel for Business Analysis

Concept: Formulas, pivot tables, charts and summarising business data. **Practice carried out:** Building SUMIFS and COUNTIFS summaries and a PivotTable of sales by category and month.

6.3 Data Collection and Cleaning

Concept: Collecting data from different sources and cleaning it for analysis. **Practice carried out:** Cleaning a data set of 200 rows with duplicates, blanks and inconsistent names and keeping a log of changes.

6.4 Descriptive Statistics

Concept: Mean, median, mode, spread and summary of business data. **Practice carried out:** Calculating the measures for twelve months of sales and interpreting the difference between the mean and the median (section 3.4).

6.5 KPI Design

Concept: Selecting measurable KPIs aligned with business goals. **Practice carried out:** Designing six KPIs with formulas, targets and owners for the store (section 3.5).

6.6 Data Visualization

Concept: Choosing charts, building graphs and presenting data clearly. **Practice carried out:** Drawing line, bar and scatter charts and correcting a misleading chart with a truncated axis.

6.7 Forecasting Basics

Concept: Trend and simple forecasting methods. **Practice carried out:** Fitting a linear trend line, calculating the forecast for month 13 and writing its limits (section 3.7).

6.8 Dashboard Building

Concept: Building a dashboard with KPIs and visuals. **Practice carried out:** Building a one-page dashboard with four KPI tiles and four charts (Figure 5).

6.9 Business Case Analysis

Concept: Analysing a business problem, options and recommendation. **Practice carried out:** Comparing three growth options for the store on cost, benefit and risk.

6.10 Insight Presentation

Concept: Presenting findings and recommendations to stakeholders. **Practice carried out:** Preparing a five-slide presentation that starts with the answer and ends with recommendations.

7. Good Practice, Ethics and Data Responsibility - Points Studied

Data and its analysis can influence important decisions, so honesty and responsibility are part of the work. The points studied are summarised below.

- **Honest presentation:** do not choose data or charts to give a false picture; show the limits and assumptions.
- **Data quality:** check the data before using it, document the cleaning steps and do not change the original.
- **Privacy:** analyse personal data only for the stated purpose and remove names where they are not needed; the Digital Personal Data Protection Act, 2023 applies.

- **Correlation and causation:** do not claim that one thing causes another unless there is a reason and evidence.
- **Reproducibility:** keep formulas visible and files organised so that another person can check the analysis.
- **Fairness:** be careful that analysis does not lead to unfair treatment of customers or employees.
- **Confidentiality:** business data and results belong to the organisation and must not be shared without permission.

8. Process Flows Studied

8.1 Steps of an analysis in Excel

- Copy the raw data into a new workbook and keep the original safe.
- Clean the data: remove duplicates, fix formats, handle blanks and standardise names.
- Add calculated columns such as profit and margin.
- Summarise with SUMIFS, COUNTIFS or a PivotTable.
- Draw charts, check them against the numbers and write the findings in words.

8.2 Choosing a chart

- Change over time - use a line chart.
- Comparison of categories - use a bar chart (horizontal for long names).
- Relationship between two numbers - use a scatter plot.
- Distribution of one number - use a histogram.
- Share of a whole with three or four parts - a pie or a stacked bar; otherwise a table.

9. Worked Examples with Sample (Dummy) Data

Worked example 1 - category performance. From the dummy data of Mahadev Mart, the sales and profit of the five categories for the year were summarised with SUMIFS, and the margin and the shares were calculated.

Category	Sales (Rs. thousand)	Profit (Rs. thousand)	Margin	Share of sales	Share of profit
Grocery	1,740	418	24.0%	38.3%	34.8%
Clothing	1,050	347	33.0%	23.1%	28.9%
Electronics	900	135	15.0%	19.8%	11.2%
Home and kitchen	570	200	35.1%	12.6%	16.7%
Stationery	280	101	36.1%	6.2%	8.4%
Total	4,540	1,201	26.5%	100%	100%

The analysis shows that grocery is the largest category but only the second-best in profit share, and that electronics, with 19.8% of the sales, gives only 11.2% of the profit because of its low margin of 15%. Clothing, home and kitchen and stationery have high margins. A possible action is to give more shelf space and promotion to clothing and home items and to negotiate better prices or reduce discounts on electronics.

Worked example 2 - month-on-month growth and the festival effect. The sales rose from Rs. 430 thousand in October to Rs. 520 thousand in November, a growth of (520 - 430) / 430 = 20.9%, and then fell to Rs. 480 thousand in December, a fall of 7.7%. The number of customers rose from 690 to 810 in the same period, while the average bill value rose from Rs. 623 to Rs. 642. The festival season therefore brings both more customers and bigger bills, and the store should plan its stock and staff for these months in advance, for example by ordering extra stock in September.

Worked example 3 - H1 and H2. The sales of the first six months were Rs. 19.8 lakh (1,980 thousand) and of the last six months Rs. 25.6 lakh (2,560 thousand), a growth of 29.3%. The profit margin rose from about 25.5% in the first half to about 27.2% in the second half, so the growth was not obtained by cutting prices. The cost as a percentage of sales fell from 74.5% to 72.8%, which suggests better buying or better control of wastage.

10. Project: Business Performance Analysis & KPI Dashboard

Problem statement. The owner of Mahadev Mart, a fictitious retail store in Gaya, wants to know how the store performed over the last twelve months, which categories and months were strong or weak, whether sales are likely to continue to grow and what should be done to improve profit. The data - monthly sales, cost and customers for twelve months and the sales and profit of five categories - is dummy data prepared for learning.

10.1 Method

- Define the questions: Which months and categories are strong? What is the margin? What drives sales? What may happen next month?
- Collect and clean the monthly and category data in Excel; check that the category totals agree with the monthly totals.
- Calculate descriptive statistics and the six KPIs.
- Draw charts, fit a trend line and calculate the forecast.
- Build the dashboard and write the findings and recommendations.

10.2 Key findings

Question	Finding
Overall performance	Sales of Rs. 45.4 lakh, profit of Rs. 12.0 lakh and a margin of 26.5%; sales grew by 29.3% in the second half.
Strong and weak months	November was the strongest (Rs. 520 thousand) and February the weakest (Rs. 295 thousand); sales are seasonal, with a coefficient of variation of 18%.
Categories	Grocery gives the highest sales; clothing and home and kitchen give high margins; electronics has the lowest margin (15%).
Drivers of sales	Sales move almost exactly with the number of customers (r = 0.9998); the average bill value rose from Rs. 596 to Rs. 632.
Forecast	The linear trend gives about Rs. 490 thousand for next January, but January is usually slow, so a range of Rs. 420 to Rs. 470 thousand is more realistic.

10.3 Recommendations

- Plan stock and staff for the festival months from September, since November sales are about 37% above the yearly mean.
- Increase the space and promotion for clothing and home and kitchen items, which together give 45.6% of the profit.
- Review the purchase prices and discounts of electronics to lift its margin from 15% towards 18%.
- Bring in more customers through offers and a loyalty card, because customers are the main driver of sales; keep encouraging bigger bills with combo offers.
- Review the dashboard monthly and compare it with the targets set in the KPI table.

10.4 Limitations

The data is dummy, covers only one year, and cannot show year-to-year seasonality. The linear forecast ignores seasonality, and the category data is assumed to be consistent with the monthly data. A real study would use at least two or three years of data, include the cost of each category fully, and test the recommendations on a small scale before applying them widely. The student noted that the learning lies in the method and the habit of checking each number.

11. Case Study Exercise: Should a Coaching Centre Open a Second Branch?

A fictitious coaching centre in Patna has 220 students in its only branch and a waiting list of 40 students. The owner is considering three options: opening a second branch in another part of the city, adding evening batches in the present branch or starting online classes. The cost of the second branch is estimated at Rs. 6 lakh as a start-up and Rs. 1.2 lakh a month, the evening batches would cost Rs. 40,000 a month, and the online classes would need Rs. 1 lakh for equipment and Rs. 30,000 a month. The student was asked to analyse the options.

Analysis. The monthly fee is Rs. 1,800 per student. The break-even number of students for each option was calculated: the second branch needs 1,20,000 / 1,800 = 67 students a month plus the recovery of the start-up cost; the evening batches need 40,000 / 1,800 = 23 students; and the online classes need 30,000 / 1,800 = 17 students plus the equipment cost. The waiting list of 40 students supports the evening batches (23 needed) immediately, and the online option is a low-risk way to reach students in other areas. The second branch has the highest potential but also the highest risk.

Recommendation. Start with the evening batches to take the waiting list, begin a small online batch at the same time and track the enrolments and the fee collection for six months. If the enrolments of the two options together exceed 90 students, consider the second branch with a better estimate of the demand. The KPIs to track are the number of enrolments, the monthly fee collection, the attendance, the dropouts and the cost per student.

Expected benefit and cautions. The recommended path uses the existing demand at a low cost and gathers the data for the larger decision. The cautions noted were that the fee and cost figures are estimates, that teacher availability may limit the evening batches, and that student data used in the analysis must be kept confidential.

12. Mock Practical Exercise: Analysing a Data Set and Presenting Recommendations

A mock practical exercise was conducted at the end of the learning phase to check that the student could carry out a complete analysis independently within a fixed time. The task was to analyse the dummy sales data of a fictitious bakery for six months and present recommendations. The steps followed were:

1. Question: the problem was stated as 'Which products and days give the most profit, and how can the profit be increased?'.
2. Data cleaning: duplicates and blanks were removed, formats corrected and a log of changes kept.
3. Summary: sales and profit by product and by weekday were summarised with a PivotTable.
4. Statistics and KPIs: the mean, median and standard deviation of daily sales and the profit margin of each product were calculated.
5. Charts: a bar chart of profit by product and a line chart of sales by weekday were drawn with clear titles.
6. Forecast: a trend line was fitted and the sales for the next week were estimated as a range.
7. Dashboard: a one-page summary with four KPIs and two charts was prepared.
8. Presentation: the findings and three recommendations were presented to the supervisor in about five minutes.

13. Sample Checklists Prepared

Data and analysis checklist

- Original data kept untouched; a working copy used.
- Duplicates, blanks, formats and units checked; every change logged.
- Formulas checked on a few rows by hand; totals agree between sheets.
- Assumptions and limits are written down.

Chart and dashboard checklist

- Chart type suits the question; title states the message; axes labelled with units.
- Bar charts start at zero; colours used with a purpose; no 3D effects.
- Dashboard fits one page; main KPIs on top; items needing attention highlighted.
- Source and date of the data are shown.

14. Tools and Resources

The following tools and resources were used during the internship: MS Excel for formulas, PivotTables, charts and trend lines, a plotting tool for the dashboard, MS Word and MS PowerPoint for the report and presentation, dummy data sets and case studies provided by the agency, templates for KPI tables and dashboards, learning material on statistics, and regular mentor feedback.

15. Programme at a Glance

The programme was in the domain of Business Analytics with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

The approximate distribution of the effort across the topics is shown below.

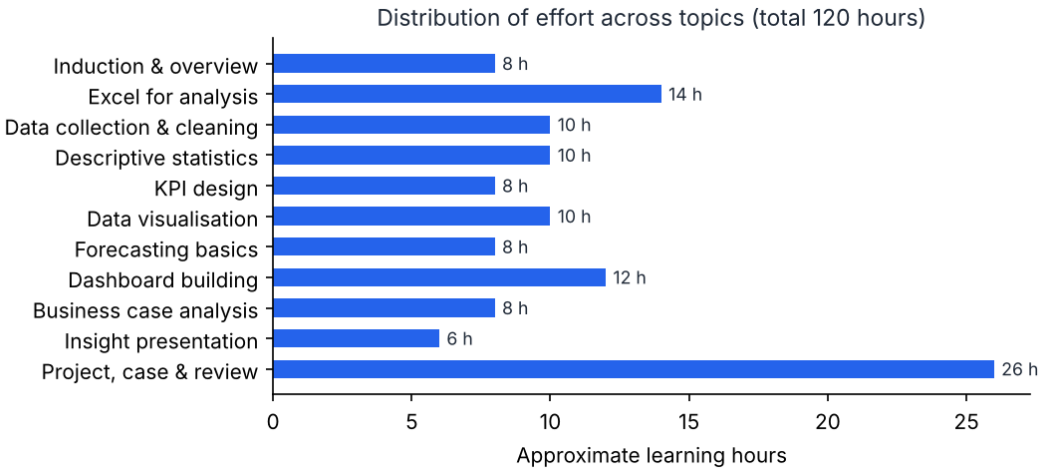


Figure 6: Distribution of learning effort across topics (120 hours)

LEARNING OUTCOME

a. Knowledge Acquired

1. The meaning, types and process of business analytics.
2. MS Excel functions, SUMIFS, COUNTIFS, lookups, PivotTables and charts.
3. Sources of business data and the cleaning of missing, duplicate and wrongly formatted values.
4. Mean, median, mode, range, standard deviation, coefficient of variation and quartiles.
5. Design of KPIs - leading and lagging - aligned with goals.
6. Choosing and building charts that tell the truth.
7. Trend lines, R-squared and the limits of forecasts.
8. Building a one-page KPI dashboard.
9. Business case analysis comparing options with cost, benefit and risk.
10. Presenting findings and recommendations to decision-makers.

Glossary of Terms Learned

- **Analytics** - Using data and methods to support decisions.
- **Descriptive / Diagnostic / Predictive / Prescriptive** - What happened / why / what may happen / what to do.
- **KPI** - Key Performance Indicator.
- **Leading / Lagging indicator** - A measure that predicts results / one that reports results.
- **Mean / Median / Mode** - Average / middle value / most frequent value.
- **Standard deviation** - The typical distance of values from the mean.
- **Coefficient of variation** - Standard deviation as a percentage of the mean.
- **Quartile** - A value that divides sorted data into four parts.
- **Outlier** - A value far from the others.
- **PivotTable** - An Excel tool to summarise data by categories.
- **SUMIFS / COUNTIFS** - Excel functions to add or count by conditions.
- **Correlation (r)** - A measure of how two numbers move together.
- **Trend line** - A line fitted to data to show the direction.
- **R-squared** - The share of variation explained by the trend.
- **Seasonality** - A pattern that repeats every year.
- **Dashboard** - A one-page display of KPIs and charts.
- **Margin** - Profit as a percentage of sales.
- **Average bill value** - Sales divided by the number of customers.
- **Stakeholder** - A person who is affected by or interested in a decision.

b. Skills Learned

Technical skills. Using MS Excel formulas, SUMIFS, COUNTIFS and PivotTables; cleaning data; calculating descriptive statistics; designing KPIs; choosing and drawing charts; fitting trend lines and making simple forecasts; building a KPI dashboard; analysing a business case; and writing and presenting findings.

Professional skills. Analytical and logical thinking; attention to detail and accuracy; curiosity and questioning; structured problem solving; clear written and spoken communication; time management; and the habit of checking and explaining every number.

These skills were developed through repeated practice and daily documentation. Working through one data set from cleaning to dashboard built accuracy and structure, checking each number built discipline, and explaining the findings to the supervisor built confidence and the ability to express results in simple words.

c. Observed Attitude and Gained Values

1. **Honesty with data:** numbers must be presented as they are, with their limits, even when the result is unwelcome.
2. **Accuracy and care:** a small mistake in data or a formula can lead to a wrong decision.
3. **Respect for privacy and confidentiality:** only dummy data was used and the importance of data protection was understood.
4. **Curiosity:** asking 'why' behind every number leads to better insight.
5. **Discipline and punctuality:** regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Understanding the difference between mean and median and when each is better was handled by calculating both for the sales data and discussing the effect of the festival months.

Using SUMIFS and PivotTables correctly was handled by checking each result against a manual total.

Cleaning the data without losing information was handled by working on a copy and keeping a change log.

Choosing a chart that makes the point clearly was handled by drawing alternatives and asking which one the supervisor understood fastest.

Understanding the limits of a trend line forecast was handled by comparing the forecast with the seasonal pattern.

Managing time between analysis, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Junior Business Analyst, Data Analyst (junior), MIS Executive, Reporting and Dashboard Executive, Sales and Marketing Analyst, Operations Analyst, Finance and Accounts Analyst and Research Assistant. Opportunities exist in banks, retail, e-commerce, telecom, healthcare, logistics, consulting firms, start-ups and government departments. Related higher-study options include MBA or PGDM in business analytics, M.Sc. in statistics or data science, BCA, MCA, and certificate courses in Excel, SQL, Power BI, Tableau and Python for data analysis. A good next step is to learn advanced Excel, SQL and one dashboard tool, practise on public data sets and keep two or three projects as a portfolio.

CONCLUSION

The internship in Business Analytics was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how business data is collected, analysed and turned into decisions - through Excel analysis, data cleaning, descriptive statistics, KPI design, visualisation, forecasting, dashboards and case analysis - and acquired practical experience by analysing the performance of a retail store and building its KPI dashboard. The experience has strengthened the student's employability and provides a foundation for further study and work in analytics and decision support.

Reflection

Looking back, the most useful lesson of the internship was that analytics is not about complicated tools but about asking a clear question, trusting only clean data and explaining the answer simply. The same table of sales told different stories when it was looked at by month, by category and by customers, and the difference between the mean and the median showed how an average can mislead. Building the dashboard showed that a good chart saves a thousand words, and that every number must lead to an action, such as stocking up before the festival or reviewing the margin on electronics.

Objectives and Achievement

The objective of understanding the types and process of analytics was achieved through the ladder of analytics and the six-step process. The objective of Excel, data cleaning and statistics was achieved through the formulas, the cleaning log and the measures of sales. The objective of KPIs and visualisation was achieved through the KPI table and the charts. The objective of forecasting and dashboards was achieved through the trend line and the one-page dashboard. The objective of case analysis and presentation was achieved through the Mahadev Mart analysis, the coaching centre case and the presentation to the supervisor.

Future Scope

The knowledge gained can support further study of data science, statistics and management, and careers in analytics, reporting and decision support. With the growth of digital payments, e-commerce and data-driven governance in Bihar and across India, organisations of every size are collecting data and need people who can turn it into clear and useful insight.

Suggestions for Improvement of the Programme

1. Provide practice with a dashboard tool such as Power BI along with Excel.
2. Include a short module on SQL for getting data from databases.
3. Add a group project using a real, public data set with permission and proper acknowledgement.
4. Continue regular mentor feedback and progress reviews.
5. Encourage students to maintain detailed daily documentation.

References

1. Kolb, D. A. (1984). Experiential Learning: Experience as the Source of Learning and Development. Prentice Hall.
2. University Grants Commission (UGC). Internship guidelines for higher education institutions.
3. Ministry of Education, Government of India. National Education Policy 2020.
4. Evans, J. R. Business Analytics: Methods, Models and Decisions. Pearson.

5. Albright, S. C. and Winston, W. L. Business Analytics: Data Analysis and Decision Making. Cengage.
6. Few, S. Show Me the Numbers: Designing Tables and Graphs to Enlighten. Analytics Press.
7. Knaflic, C. N. Storytelling with Data. Wiley.
8. Levine, D. M. et al. Statistics for Managers Using Microsoft Excel. Pearson.
9. Government of India. The Digital Personal Data Protection Act, 2023.
10. RK NEXORA internship programme handbook.

ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]